

Task 1

A) Data Clean

```
import pandas as pd
album_data = pd.read_csv("data/Top_10_Albums_By_Year.csv")
anime_data = pd.read_csv("data/Anime.csv")
algorithm_data = pd.read_csv("data/algorithm_trials.csv")

print("Displaying Album Data Columns")
print(album_data.columns)
print("\n\nDisplaying Anime Data")
print(anime_data.columns)
print("\n\nDisplaying Algorithm Data")
print(algorithm_data.columns)
```

Displaying Album Data Columns

```
Index(['Year', 'Ranking', 'Artist', 'Album', 'Worldwide Sales (Est.)',
      'Tracks', 'Album Length', 'Genre'],
      dtype='object')
```

Displaying Anime Data

```
Index(['Rank', 'Name', 'Japanese_name', 'Type', 'Episodes', 'Studio',
      'Release_season', 'Tags', 'Rating', 'Release_year', 'End_year',
      'Description', 'Content_Warning', 'Related_Mange', 'Related_anime',
      'Voice_actors', 'staff'],
      dtype='object')
```

Displaying Algorithm Data

```
Index(['Unnamed: 0', 'Epoch', 'Algorithm', 'Run', 'Accuracy'], dtype='object')
```

Above are all of the columns associated with each of the datasets

B) Generate Visualizations

C) Justify Best Visualization

Task 2

A) Develop Hypothesis

B) Fitting Distributions

C) Comparing Real and Synthetic Data